Smart Expense Tracker and Goal-Based Saving Analyzer

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***Abstract*—Managing personal finances requires more than recording purchases: people need to see where their money goes, check progress toward savings goals, and revise plans when income or expenses change. This paper presents the design of the Smart Expense Tracker and Goal-Based Saving Analyser (SET-GBSA). This web-based tool integrates transaction recording, category-wise analytics, goal tracking and explainable advice. Users record income and expenses, define a savings goal with a target amount and deadline, and monitor progress on a dashboard of bar, doughnut, line and progress charts. The system computes the monthly contribution needed to reach a goal and, when a contribution is missed, recalculates the requirement while checking it against the user's estimated surplus. A moving-average baseline forecasts short-term category expenses, and a rule-based module produces recommendations phrased as options rather than commands. The paper describes the architecture, data model, methodology, implementation design and a worked numerical example, and defines an evaluation framework covering functional correctness, forecast error, goal-plan accuracy and usability. No empirical results are claimed; the worked example is illustrative. SET-GBSA is intended as an educational aid to financial awareness, not as a substitute for professional financial advice.**

***Index Terms*—Expense tracking, goal-based saving, personal finance, transaction analytics, expense forecasting, adaptive planning, data visualisation.**

I. INTRODUCTION

Every day, people make financial choices that appear in purchases, bills, subscriptions and savings promises. When records are kept in scattered applications or reviewed only at month-end, it becomes hard to spot spending patterns or to see whether saving habits match future plans. A plain ledger records how much money was spent; it does not show where spending is shifting, how much should be saved next month, or what to do after a missed saving [1], [2].

Digital payments make transaction histories easier to inspect, and data analytics can turn those histories into summaries and short-term predictions. Recent work such as Smart Pocket [7] shows that machine-learning-based categorisation and visual dashboards can give users clearer visibility of their spending. A practical personal-finance tool, however, must combine such analysis with clear explanations and real user control: a prediction without context can confuse, and a savings goal that never adapts may feel unreachable when circumstances change.

This paper presents the Smart Expense Tracker and Goal-Based Saving Analyser (SET-GBSA). SET-GBSA brings together transaction recording, category-wise analysis, savings-goal management, progress visualisation and explainable advice on a single screen. Its central idea is to link a goal timeline with cash-flow analysis: if a user saves less than planned, SET-GBSA recalculates the remaining amount and time and checks whether the revised contribution fits within an estimated budget.

The paper offers four contributions: (1) a modular design that merges expense tracking with goal planning; (2) a transparent method for recalculating savings after a slip; (3) a mixed analytics approach that combines simple descriptive statistics, baseline forecasting and easy-to-understand rules; and (4) an evaluation framework that can be applied to a working prototype.

The remainder of the paper is organised as follows. Section II reviews related work, Section III states the problem and objectives, Section IV describes the system architecture, Section V the methodology, Section VI the implementation design and Section VII the evaluation framework. Section VIII gives an illustrative worked example, followed by the discussion, future scope and conclusion.

II. RELATED WORK

Personal money tools typically provide transaction lists, budgets, category summaries and charts. These features help people see what happened in the past, but their usefulness depends on consistent data entry, meaningful categories and the user's ability to turn observations into action. Studies of financial skills and saving behaviour indicate that clear information and feedback help, although the real effect of any tool depends on who uses it and for how long [1], [2].

Budgeting systems normally compare spending with a set limit. Goal-based systems add a target amount and a deadline, allowing progress to be expressed as a percentage or a projected completion date. A static plan can fail when a user misses a saving or when expenses change. A flexible plan can recalculate the required saving rate, but it must not present a single number as if it were always achievable.

Smart Pocket [7] is the closest reference point for this work. It classifies expenses with a Random Forest model, forecasts spending with an LSTM network, and presents category statistics, a monthly budget gauge and time-series charts on a dashboard; it reports a classification accuracy of 93.4% on 1,250 transactions from 27 users. Its reported features centre on classification, budget monitoring and forecasting, and a goal-and-deadline planning component is not described. SET-GBSA keeps the dashboard-oriented presentation of [7] but shifts the emphasis to goal-based saving, adaptive recalculation and explainability, as summarised in Table I.

TABLE I
COMPARISON WITH THE BASE WORK

Feature	Smart Pocket [7]	SET-GBSA (proposed)
Primary aim	Expense classification, budget monitoring, spending prediction	Goal-based saving linked to expense analysis
Categorisation	Random Forest classifier	User-confirmed, keyword-assisted categories
Forecasting	LSTM network	Moving-average baseline (smoothing optional)
Progress view	Monthly budget gauge	Goal progress and required contribution
Missed saving	Not described	Recalculation with affordability check
Evaluation	Results reported in [7]	Framework defined; no results claimed

Time-series forecasting methods such as moving averages and exponential smoothing can estimate future values from past observations [3], [4]. Advanced machine-learning models can capture hidden patterns, but they usually need more data and are harder to explain. For a prototype, a simple baseline forecast with clear rules is therefore preferable to a complex model whose predictions are hard to justify. SET-GBSA accordingly separates analytics, forecasting and recommendation logic into independent components so that each can be tested on its own.

The design also follows transparency and data-minimisation principles. Users must know what data is stored, why a recommendation appears and how to correct a transaction. This matters because financial records are sensitive: the system should collect only the data it needs, keep login credentials safe and never initiate automatic money movement.

III. PROBLEM STATEMENT AND OBJECTIVES

The problem is the gap between keeping track of money and having a savings plan. Users need one view that shows cash flow, spending categories, goal progress and the monthly amount needed to meet a deadline. The system must also cope with missing records, uneven income and skipped saving months, without pretending to know the future. The objectives are to:

- provide user access and keep a clear record of income and expenses;
- sort transactions into easy-to-understand categories and summarise them each month;
- show income, spending, net balance, savings and category trends on charts;
- let the user set a goal amount, current savings, target date and planned monthly contribution;
- recalculate the needed contribution if a month is missed or only partly saved, and show the assumptions used;
- predict short-term expenses when enough history exists and make the uncertainty obvious; and
- give clear recommendations and alerts while leaving the final choice to the user.

IV. PROPOSED SYSTEM ARCHITECTURE

SET-GBSA uses a modular client–server structure, shown in Fig. 1. The presentation layer offers registration, login, transaction entry, goal entry and a dashboard. The application layer checks that input follows business rules, computes summaries and manages forecasting. The data layer keeps user profiles, transactions, goals, contributions and optional prediction logs. A chart component turns aggregated numbers into bar, line, doughnut and progress visuals.

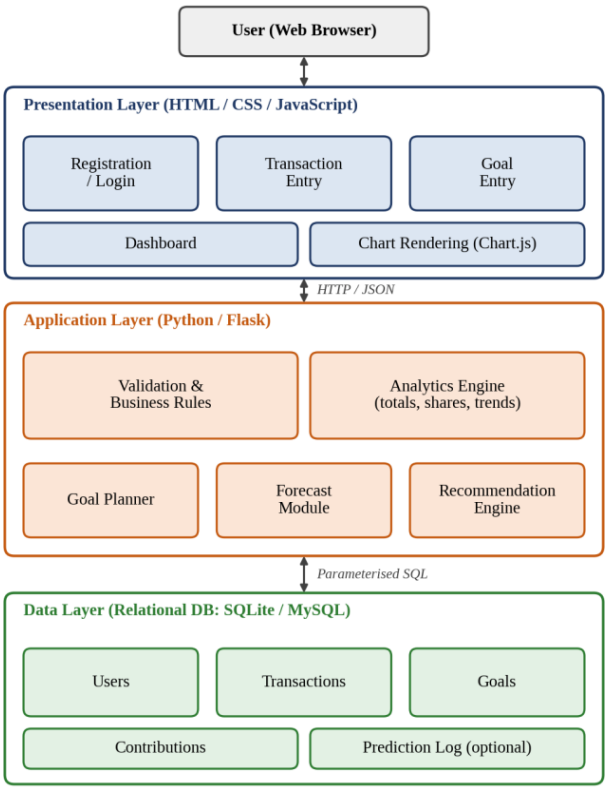


Fig. 1. Layered architecture of SET-GBSA.

The application layer contains five modules. The *validation and business-rules* module checks each entry before it is stored. The *analytics engine* computes totals, category shares and trends. The *goal planner* calculates remaining amounts and required contributions. The *forecast module* estimates

short-term category expenses. The *recommendation engine* turns the results into observations and options.

A typical flow starts when the user logs in and adds transactions. The system validates the entries and saves them for that user, the analytics engine gathers the data and the dashboard is refreshed. When the user creates or changes a goal, the planning module works out how fast saving must proceed. At the end of a month, or when the user logs a contribution, the system compares progress with the plan and updates the remaining plan. Forecasts and suggestions appear when enough data exists.

V. METHODOLOGY

The method has five steps: capture the data, validate and categorise it, analyse it, calculate the goal plan, and give feedback (Fig. 2). Each step is detailed below.

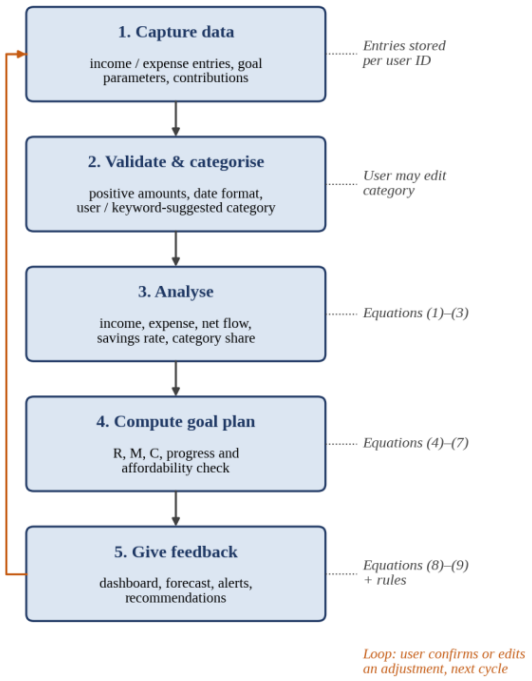


Fig. 2. Five-step methodology workflow.

A. Data Collection and Data Model

In contrast to [7], which drew on receipts, bank statements and payment notifications, the present design of SET-GBSA works only with data entered by the user: income, expenses, goals and contributions. No dataset was collected for this paper. A relational database holds the objects (Fig. 3). A *User* table keeps an ID, an email or name and a securely hashed password. A *Transaction* table keeps a transaction ID, the user ID, the date, the amount, whether it is income or expense, a category and an optional note. A *Goal* table keeps a goal ID, the user ID, the goal name, the target amount, the amount already saved, the start date, the deadline and the status. A *Contribution* table records deposits for a goal, so that real contributions can be told apart from ordinary balances. Foreign-key links tie each record to its owner, and every query must be restricted to the authenticated user's data.

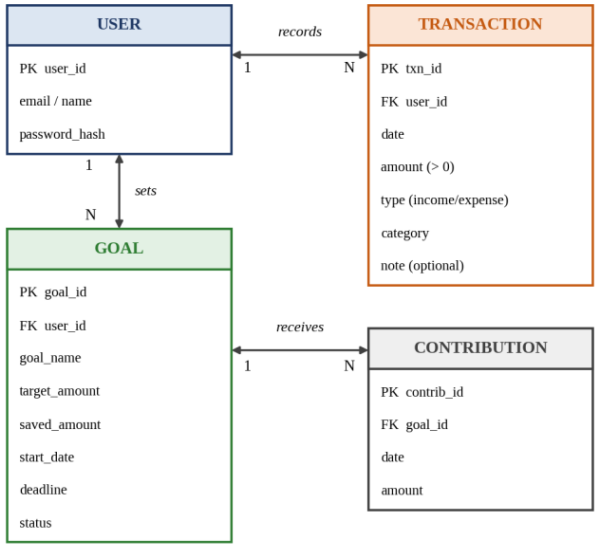


Fig. 3. Entity–relationship diagram of the data model.

B. Data Validation and Categorisation

Every amount must be positive; the type field says whether the number is income or spending. Dates are stored in a single standard format. The user can pick a category, or the system may suggest one using keywords in the note; the user can correct any category the system chooses. Entries that fail validation are rejected with an understandable message.

C. Descriptive Financial Measures

For a chosen period, total income I is the sum of all income entries and total expenses E is the sum of all expense entries. Net cash flow is

$$N = I - E \tag{1}$$

If savings contributions are kept separate, the system must not count them as expenses; it labels them as transfers toward a goal. When income is positive, the savings rate is

$$SR = (\text{amount saved} / I) \times 100 \tag{2}$$

and, when total expenses are not zero, the share of a category is

$$CS = (\text{category expense} / E) \times 100 \tag{3}$$

These metrics support comparisons and trend charts.

D. Goal Progress and Adaptive Contribution

Let G be the target amount, S the amount already saved toward the goal and M the number of contribution periods left until the deadline. The remaining amount is

$$R = \max(G - S, 0) \tag{4}$$

When M is greater than zero, the baseline contribution required per period is

$$C = R / M \tag{5}$$

and, for G greater than zero, the progress percentage is

$$P = min[(S / G) \times 100, 100] \quad (6)$$

If a user misses a contribution, S remains unchanged; the application recalculates R and M using the current date, so the revised contribution is higher when time has passed without the expected saving.

The calculation is mathematically straightforward, but affordability needs additional context. Let A be the user's estimated surplus for the next period, based on expected income minus essential and planned expenses. The shortfall is

$$D = max(C - A, 0) \quad (7)$$

If D > 0, the interface does not silently prescribe the amount. It shows the shortfall and offers user-controlled options: extend the deadline, reduce the target, make an optional additional contribution, or review discretionary expenses. Any adjustment must be confirmed by the user, which avoids presenting a goal schedule as a guarantee. The complete decision logic is shown in Fig. 4.

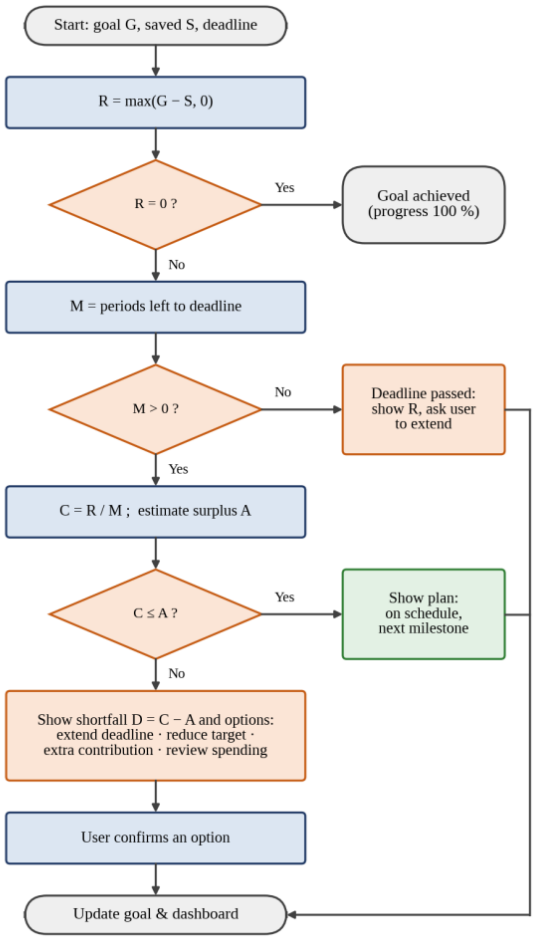


Fig. 4. Flowchart of goal progress, adaptive contribution and affordability check.

E. Expense Forecasting

A baseline forecast uses a moving average of the k most recent complete months for each category:

$$F = (e1 + e2 + \ldots + ek) / k \quad (8)$$

where ei is the category total for the i-th most recent month. The parameter k is configurable and should be chosen through validation. Exponential smoothing is an alternative when recent months should receive more weight [3], [4]:

$$Ft+1 = \alpha\, yt + (1 - \alpha)\, Ft, \quad 0 < \alpha \leq 1 \quad (9)$$

For short histories, the system displays a data-insufficient message or uses a simple average rather than producing a misleadingly precise prediction.

F. Recommendation Logic

Recommendations are first generated through rules (Table II). Each is phrased as an observation and an option rather than a command, and states the data period used.

TABLE II
RULE-BASED RECOMMENDATION LOGIC

Rule	Trigger	Output
Category rise	Category total exceeds its recent average by threshold θ	Flag the increase and show the underlying amounts
Unaffordable plan	Required contribution C exceeds surplus A	Explain shortfall D and offer adjustment options
On schedule	Savings meet or exceed the planned pace	Show the pace and the next milestone
Short history	Fewer than k complete months of data	Show a data-insufficient notice or a simple average

VI. IMPLEMENTATION DESIGN

A prototype can be implemented with Python and Flask for server-side routing, HTML and CSS for the interface, JavaScript for interactive components and SQLite or MySQL for persistence. A charting library such as Chart.js can render expense bars, category-wise doughnut charts and goal-progress indicators. Python libraries such as Pandas and scikit-learn may support data preparation and optional forecasting experiments. The stack is summarised in Table III and can be adapted to requirements.

TABLE III
PROPOSED TECHNOLOGY STACK

Layer	Technology	Role
Server	Python, Flask	Routing, services, business rules
Interface	HTML, CSS, JavaScript	Forms, dashboard, interaction
Charts	Chart.js	Bar, doughnut, line and progress visuals
Database	SQLite or MySQL	Users, transactions, goals, contributions
Analysis	Pandas, scikit-learn (optional)	Data preparation, forecasting experiments

The backend exposes routes or services for authentication, transaction operations, goal operations, dashboard summaries and forecasts. Input validation must occur on the server even when client-side validation is present. Passwords must be hashed with a recognised password-hashing function; plain-text passwords are never stored. Database

queries are parameterised to reduce injection risk [6], and error messages are understandable without revealing internal details.

The interface prioritises a small set of indicators: total income, total expenses, current savings, goal completion percentage and the contribution required for the next period. Each chart has a title, date range, units and an accessible text summary. Users can filter by month, edit transactions, export their own records and delete their account. Financial data is not shared with other parties without informed consent.

VII. EVALUATION FRAMEWORK

The proposed system should be evaluated using analytical and usability tests, summarised in Table IV. Functional testing checks registration, login, transaction creation and editing, category aggregation, goal creation, contribution recording and recalculation after a missed period. Boundary tests should include a zero target, a past deadline, no transaction history, zero income, a contribution larger than the remaining goal and a target already achieved.

TABLE IV
PROPOSED EVALUATION MEASURES

Aspect	Measure	Method
Functional	Test-case pass rate	Registration to recalculation; boundary tests
Forecasting	MAE, RMSE	Chronological holdout vs. baselines
Goal plan	Agreement with hand-verified values	Worked examples for R, M, C
Usability	Completion rate, time on task	Task study and questionnaire
Advice	Clarity and relevance ratings	Participant rating of each explanation

Forecasting performance can be measured using the mean absolute error (MAE) and root mean squared error (RMSE) on a chronological holdout set, where y_i is the observed and $\hat{y}_i$ the forecast value over n test months and the sums run over $i = 1, \ldots, n$:

$$MAE = (1/n) \sum |y_i - \hat{y}_i| \quad (10)$$

$$RMSE = \sqrt{[(1/n) \sum (y_i - \hat{y}_i)^2]} \quad (11)$$

Randomly shuffling time-series records should be avoided because it can leak information into training. Forecasts should be compared with baselines such as the previous month's value and a historical average, and results should be reported separately for categories with sufficient observations.

Goal-plan accuracy can be assessed by checking the remaining amount and contribution against hand-verified examples. Usability can be studied through task completion rate, time on task and a short user questionnaire [5]. Recommendation usefulness can be assessed by asking participants whether each explanation is understandable and relevant. A future study should recruit participants with budgeting habits and clearly report sample size, study duration and limitations. **No empirical results are claimed in this design paper.**

VIII. ILLUSTRATIVE ANALYSIS AND RESULTS

To show how the measures behave, this section applies the equations to a hypothetical user. The values are invented for illustration only; they are not experimental results, and no user study has been conducted.

A. Category Analysis

Suppose the user has a monthly income I = ₹30,000 and total expenses E = ₹25,000. By (1), the net cash flow is N = ₹5,000. The category totals in Fig. 5 give shares by (3) of 36.0% for Rent and 28.8% for Food. If the user contributes ₹4,500 to a goal in the month, the savings rate from (2) is 15%. Shopping (₹2,100) is 23.5% above its recent average of ₹1,700; with a threshold θ = 15%, the recommendation module would flag it and show both amounts.

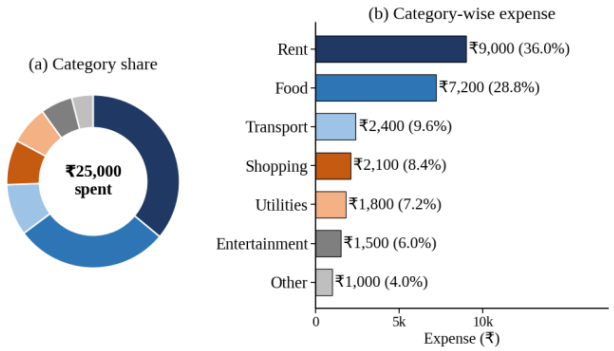


Fig. 5. Illustrative category analysis: (a) share of total expenses, (b) category-wise expense.

B. Goal Plan and Missed Contribution

Let the goal be G = ₹60,000 with S = ₹15,000 saved and M = 10 months to the deadline. Then R = ₹45,000 by (4) and C = ₹4,500 by (5). Table V traces the plan: the user saves ₹4,500 in month 1 but misses month 2. Progress at the end of month 2 is P = 32.5% by (6), eight months remain and the revised contribution rises to ₹5,062.5 (≈ ₹5,063), as shown in Fig. 6.

TABLE V
WORKED EXAMPLE OF ADAPTIVE RECALCULATION

Month	Saved (₹)	S (₹)	M	R (₹)	C (₹)	Progress
0	–	15,000	10	45,000	4,500	25.0%
1	4,500	19,500	9	40,500	4,500	32.5%
2	0 (missed)	19,500	8	40,500	5,063	32.5%

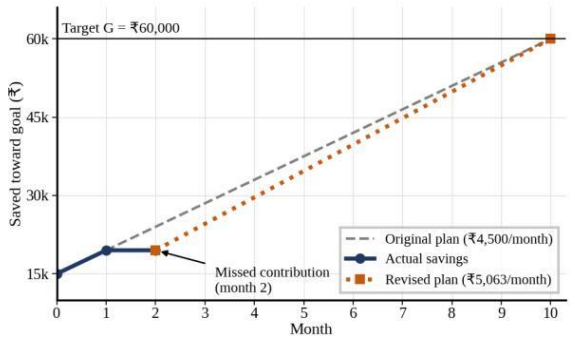


Fig. 6. Illustrative goal trajectory showing the original plan, actual savings and the revised plan after a missed contribution.

Suppose the user's estimated surplus is A = ₹4,200. By (7), the shortfall is D = ₹862.5 > 0, so the system shows the shortfall and the adjustment options instead of prescribing the higher amount. One option, extending the deadline, requires $M \geq 40{,}500 / 4{,}200 \approx 9.64$, that is, 10 periods or two additional months. The user must confirm this option before the goal is updated.

C. Baseline Forecast

For the Food category, six months of history are shown in Fig. 7. With k = 3, (8) gives F = (6,900 + 7,500 + 7,200) / 3 = ₹7,200 for the next month. The dashboard labels this value as an estimate and states the months used (M4–M6). With fewer than three complete months, it would display the data-insufficient notice. These examples demonstrate the computations and their presentation, not the performance of the system.

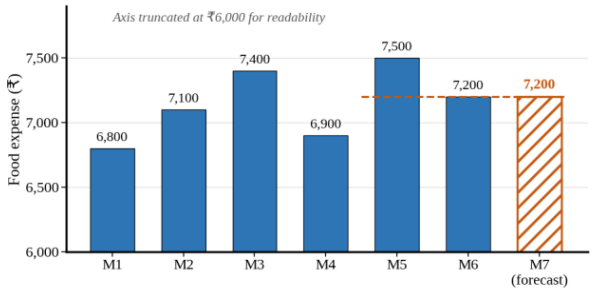


Fig. 7. Illustrative three-month moving-average forecast for the Food category.

IX. DISCUSSION

The main practical value of SET-GBSA is the connection between spending analysis and forward-looking goal planning. Charts make patterns easier to inspect, while recalculation helps users understand the consequences of a missed contribution. Separating the plan from the affordability check is important: the amount required to meet a deadline may not match what a user can actually afford given their cash flow.

Several limitations remain. Transaction data may be missing, misclassified or unrepresentative of true spending. Irregular income and unexpected expenses can make forecasts less reliable, and a moving average cannot anticipate one-off events. Rule-based recommendations may not fit every user's situation, and the prototype relies on users recording transactions consistently, which remains a challenge without an authorised integration. Forecasts should therefore be clearly labelled as estimates, recommendations must be explainable, and users should always keep control over any change.

X. FUTURE SCOPE

Future improvements could include integration with banks or payment providers, but only with clear user consent. Better category classification would help; once labelled data is available, machine-learning classifiers such as the Random Forest approach reported in [7] could be compared against the keyword-assisted baseline. Uncertainty intervals on forecasts would give a more honest picture, and recurring-bill detection could improve planning. User studies over several months would help test long-term usability. Any real integration with payment systems would require security, legal and operational checks. The project should remain focused on awareness and planning: it should not handle payments or make investment decisions.

XI. CONCLUSION

This paper presented the design of a Smart Expense Tracker and Goal-Based Saving Analyzer. It combines transaction management, visual spending analysis, goal monitoring, adaptive contribution calculations and explainable recommendations. The core idea is to link spending history with a user-defined future goal while keeping all assumptions and limits visible. A modular setup using a web framework, a relational database and charting tools can support a student-level prototype, and the evaluation framework provides a way to test correctness, forecasting quality and usability without overstating results that have not been measured. By keeping recommendations transparent and under user control, the system can serve as an educational tool that improves financial awareness.

REFERENCES

[1] OECD, OECD/INFE 2020 International Survey of Adult Financial Literacy. Paris, France: OECD Publishing, 2020.

[2] Consumer Financial Protection Bureau, Building Your Savings: A Report on the Use of Savings Strategies. Washington, DC, USA: CFPB, 2020.

[3] R. J. Hyndman and G. Athanasopoulos, Forecasting: Principles and Practice, 3rd ed. Melbourne, Australia: OTexts, 2021.

[4] R. J. Hyndman, A. B. Koehler, J. K. Ord, and R. D. Snyder, Forecasting with Exponential Smoothing: The State Space Approach. Berlin, Germany: Springer, 2008.

[5] A. Field, Discovering Statistics Using IBM SPSS Statistics, 5th ed. London, U.K.: SAGE Publications, 2018.

[6] OWASP Foundation, OWASP Top 10: The Ten Most Critical Web Application Security Risks, 2021.

[7] B. Mahto, A. K. Varma, S. Kumari, R. Parveen, C. Firdous, A. Sharma, and K. Amrendra, “Smart Pocket: A machine learning–based expense tracker and spending predictor,” Int. J. Res. Innov. Appl. Sci. (IJRIAS), vol. X, no. XI, pp. 352–362, Nov. 2025, doi: 10.51584/IJRIAS.2025.101100032.